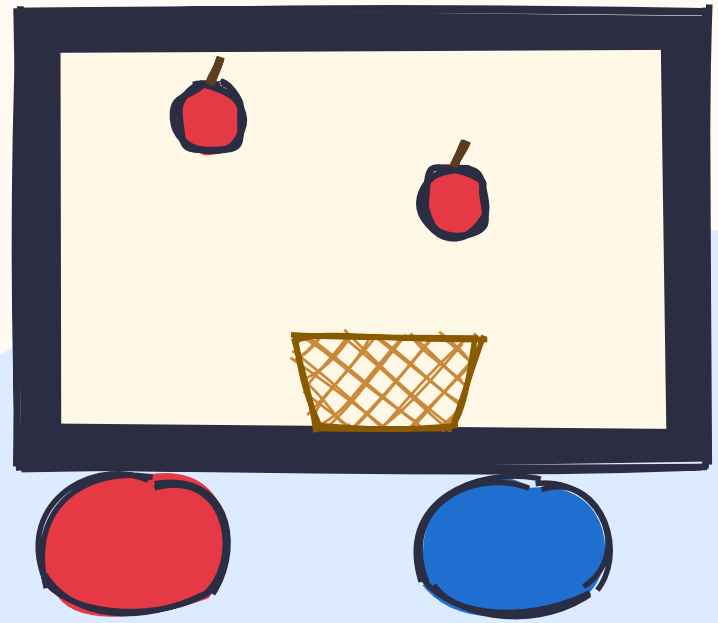
Little Explorers · Class 15

Catch-the-Fruit Game

Make a game AND the controller you play it with!

Playdough buttons **Scratch game** **Game testing**



Did you know?
Game companies hire 'game testers' whose job is to play games all day and find problems!

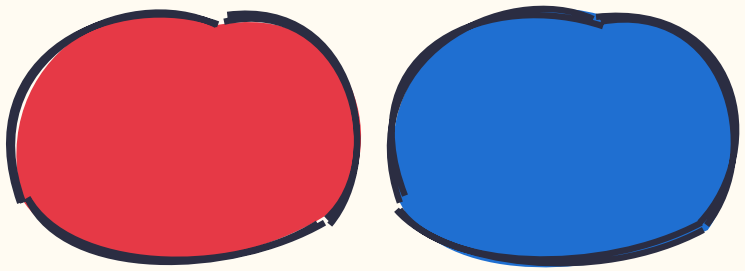
LEARN

Let's wonder...

Can playdough control a game?

A squishy surprise

Game controllers have buttons and sticks. What if we made our **own** controller from playdough? Would it work?



Too squishy!

Maybe!

Let's build it!

Today: we build a **playdough controller** and use it to play a fruit-catching game!

Learning objectives

Today I can...

1	**Controller:** build a **playdough controller** with left and right buttons	**Proof:** pressing my dough moves the basket
2	**Game rules:** change the **speed** and **score** in a Scratch game	**Proof:** I changed one rule and it worked
3	**Test:** let a **friend test** my game and improve it	**Proof:** I changed something after my friend played

Objective 1 · Controller · video

Video time!



Hands-on Science with Squishy Circuits

TED (AnnMarie Thomas) · about 4 minutes · youtu.be/5M3Dow20KlM

Watch for...

1. What is the special playdough made from?
2. What can you light up with playdough?
3. Why can playdough carry electricity?

Scan to watch

After the video: Squeeze your hands like playdough. Tell your partner one thing you would build with squishy circuits.

Objective 1 · Controller

Salty playdough is a conductor

The idea

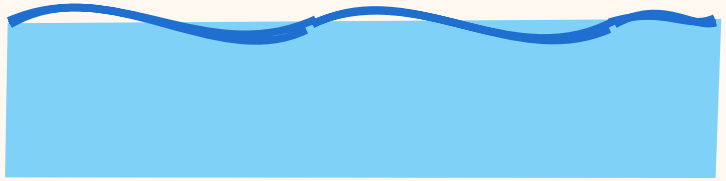
Our special playdough has **salt** and **water** in it. Salty water lets a tiny electricity flow, so the dough is a **conductor**!

We clip the dough to the Makey Makey's **left** and **right** arrow keys. Touch the dough = press the key!

Not food
Do not eat the playdough. Wash your hands after playing!

Example: the sea

salty = conducts!



Sea water is salty and conducts electricity much better than pure rainwater. Our playdough is a little bit like squishy sea water!

Objective 2 · Game rules

Game rules are just code

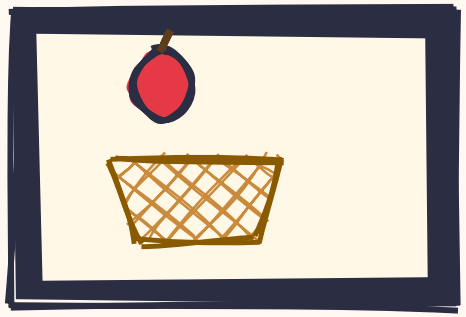
The idea

Our Scratch game has rules written in blocks:

- **when left arrow pressed** → move basket left
- **fruit falls down** at a **speed**
- **if** fruit touches basket → **score + 1**

Change a number and the game changes!

Example: playground tag



score: 3

In tag, the rule might be: 'If I touch you, you are it!' Change the rule to 'touch with two hands' and the game feels different. Game rules are the same in code!

Real world

Controllers come in all shapes



Game pads
Buttons and sticks



Motion controllers
Swing your arm to play tennis



Adaptive controllers
Big buttons for players with disabilities



Touch screens
Tap and swipe

? If you could invent a new controller, what would it be made of?

Inventors
Some controllers are designed so everyone can play, including people who cannot use their hands well.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Build the controller
2. Human catch!
3. Remix the game
4. How fast is fun?
5. Be a game tester
6. Game designer challenges

Objective 1 · Build step 1 of 2

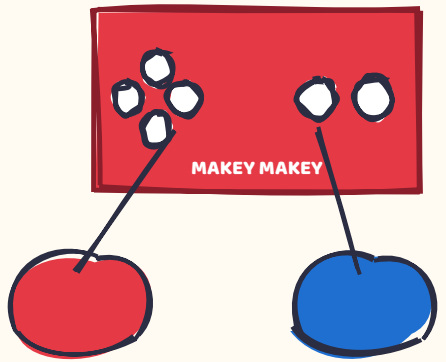
Build the controller

a Roll two balls of playdough: one **red** (left), one **blue** (right).

b Clip the red ball to the **left arrow**, the blue ball to the **right arrow**.

c Clip a wire to **EARTH** and hold it (or wear a foil bracelet on it).

d Press the red dough: does the basket move left?



Check: Red dough moves the basket left, blue dough moves it right.

Keep the two dough balls apart. If they touch, both keys press at once!

Class game

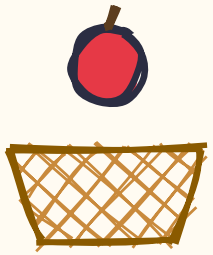
Human catch!

How to play

1. Two children hold a big basket (or a box) between them.
2. Another child gently drops soft balls from a chair.
3. The class shouts **"Left!"** or **"Right!"** to help.
4. Count how many balls they catch in 30 seconds.
5. Swap players!

Then discuss

Was it easier when the balls fell **slowly** or **fast**? Should our game get faster?



Big idea

In every game, the rules decide how hard or easy it is.

MAKE

Objective 2 · Build step 2 of 2

Remix the game

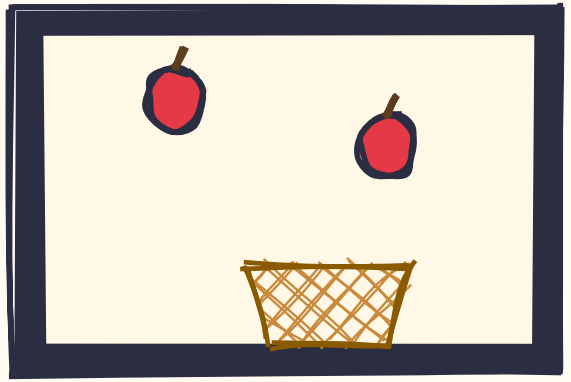
a Open the class **Catch-the-Fruit** project in Scratch.

b Find the block **change y by -3**. Make it **-5**: faster fruit!

c Find **change score by 1**. Make it **2**: double points!

d Click a new **sound** for when you catch a fruit.

e Play with your playdough controller!



Check: The game is faster, scores more and plays your new sound.

Test it!

How fast is fun?

Fruit speed	It felt…	My score
Speed -2	too slow / just right / too fast?	_____
Speed -5	too slow / just right / too fast?	_____
Speed -10	too slow / just right / too fast?	_____

Talk about it

Is the most fun game the **easiest** one? Why or why not?

Same time

Play for 30 seconds each time so it is fair.

Objective 3 · Test

Be a game tester

How to test

Swap seats with a friend. Play their game for **1 minute**. Then tell them:

⭐ one thing you **loved**

💡 one thing to **change**

Feedback card	
I loved…	_____
Please change…	_____
My change after testing	_____

Kind words

Say 'I liked it when…' and 'Maybe you could…'

Level up!

Game designer challenges

Choose your challenge



Golden fruit: a special fruit worth 5 points!



Bad apple: catch a rotten apple and lose a point.



Speed up: the fruit falls faster after every 5 points.



Jump button: add a third dough ball for the space key.

Think like an engineer

Why do games get harder as you play?

Boss level

Make a two-player game with two playdough controllers!

Objective check

Can I really do it? Prove it.

01	Show your playdough controller moving the basket left and right.	○ ○ ○
02	What rule did you change in the game? What happened?	○ ○ ○
03	What did your friend say, and what did you change?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Game arcade: half the class plays while the other half tests, then swap.

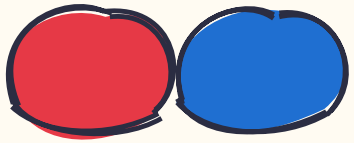
Show what you know

What did we learn?

- [x] Salty playdough is a conductor
- [x] Dough + Makey Makey = a homemade controller
- [x] Game rules are code: change a number, change the game
- [x] Testers help make games better

At home

Tell your family the rules of your game. Can they find a way to make it better?



Exit ticket: what happens if the two dough balls touch?

Next class: play the Robot Sorting Rules Game!